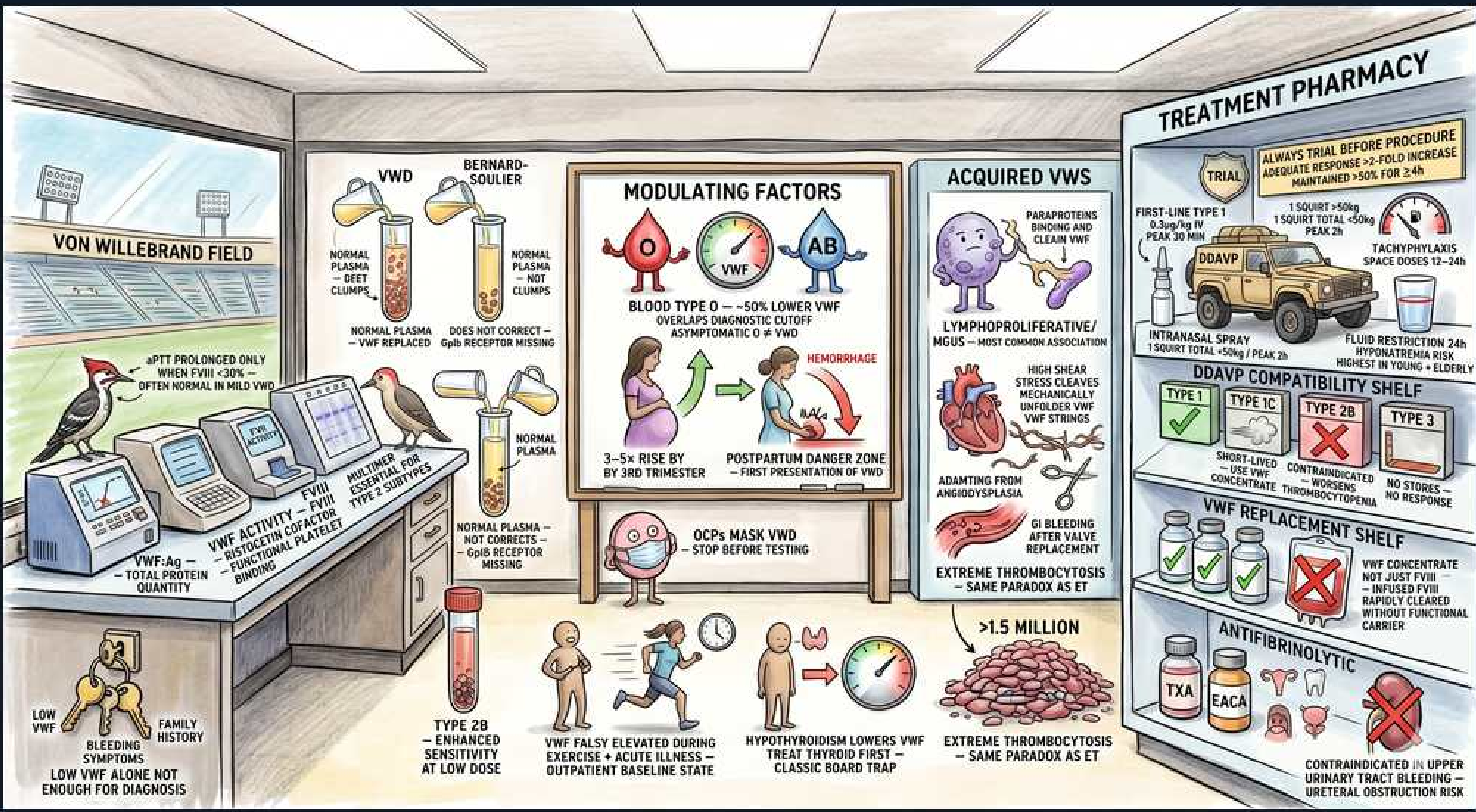
Platelet disorders — von Willebrand disease treatment



1. aPTT prolonged in severe

WHAT YOU SEE

The lab bench note 'aPTT PROLONGED ONLY WHEN FVIII LOW <30%, OFTEN NORMAL IN MILD VWD' — a normal aPTT does not clear mild vWD.

WHAT IT ENCODES

The aPTT in vWD is prolonged ONLY when vWF falls enough to drop carried Factor VIII below ~30-40%. In mild type 1 the FVIII is often normal, so the aPTT is NORMAL — meaning a normal aPTT does NOT exclude vWD. PT is always normal. The definitive panel is vWF:Ag, vWF activity (ristocetin cofactor / vWF:GpIbM), and FVIII level; PFA-100 closure time is the modern screen for primary-hemostasis defects.

BOARD TRAP

WRONG: ruling out vWD because the aPTT and PT are both normal. Mild vWD frequently has a completely normal coagulation profile. Discriminator: order vWF antigen + activity directly when there is mucocutaneous bleeding or menorrhagia — do not rely on aPTT as a screen.

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2. Low vWF ≠ diagnosis alone

WHAT YOU SEE

The bottom-left keys/note 'LOW vWF + BLEEDING SYMPTOMS + FAMILY HISTORY' captioned 'LOW vWF ALONE NOT ENOUGH FOR DIAGNOSIS' — all three pieces required.

WHAT IT ENCODES

Diagnosis of vWD requires the TRIAD of (1) low/dysfunctional vWF on labs, (2) a personal bleeding history (mucocutaneous bleeding, menorrhagia, post-surgical bleeding — quantified by a bleeding assessment tool), and (3) family history. 'Low vWF' (levels ~30-50 IU/dL) without bleeding is now often labeled a risk factor rather than disease. Levels are dynamic — repeat testing is often needed because vWF is an acute-phase reactant.

BOARD TRAP

WRONG: diagnosing vWD off a single borderline-low vWF level in an asymptomatic patient. vWF rises with stress, exercise, inflammation, estrogen, and pregnancy, so one value can mislead either way. Discriminator: require bleeding symptoms + repeat/confirmatory testing before labeling someone with vWD.

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3. Blood group O lowers vWF

WHAT YOU SEE

The chalkboard blood-drop labeled type O against type AB with the gauge showing '~50% LOWER vWF / ASYMPTOMATIC = NORMAL O' — group O physiologically runs vWF low.

WHAT IT ENCODES

ABO blood group is the largest genetic determinant of vWF level: type O individuals have vWF roughly 25-50% LOWER than non-O groups because the O glycan accelerates vWF clearance. Many labs use blood-group-adjusted reference ranges. This is why a mildly low vWF in a type O person with no bleeding is frequently a normal variant rather than true disease.

BOARD TRAP

WRONG: diagnosing type 1 vWD in an asymptomatic type O patient with a mildly low vWF level. Discriminator: in blood group O, a slightly low vWF can be a physiologic normal variant — correlate with bleeding history and use blood-group-adjusted ranges before diagnosing.

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4. Pregnancy raises vWF

WHAT YOU SEE

The pregnant-figure timeline '3-5x RISE BY 3RD TRIMESTER' leading into a 'POSTPARTUM DANGER ZONE / FIRST PRESENTATION OF VWD' — vWF climbs in pregnancy then crashes after delivery.

WHAT IT ENCODES

vWF and FVIII rise PROGRESSIVELY in pregnancy (estrogen-driven), reaching ~3-5x baseline by the third trimester. Many type 1 patients normalize their levels and bleed minimally at delivery. The DANGER ZONE is POSTPARTUM: levels fall sharply over days 1-3 and can precipitate delayed primary or secondary postpartum hemorrhage — monitor vWF/FVIII and continue prophylaxis (DDAVP or concentrate) for several days after delivery.

BOARD TRAP

WRONG: stopping vWD precautions right after a smooth vaginal delivery because peripartum levels were normal. Discriminator: the postpartum vWF crash (days 1-3 to ~2 weeks) is the high-bleeding window — keep monitoring and treating after delivery, not just during it.

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5. OCPs mask vWD

WHAT YOU SEE

The OCP pill-pack figure captioned 'OCPs MASK VWD / STOP BEFORE TESTING' — estrogen falsely normalizes the labs.

WHAT IT ENCODES

Estrogen-containing oral contraceptives RAISE vWF and FVIII and can normalize labs, masking vWD. Conveniently, this also makes combined OCPs a useful TREATMENT for menorrhagia in known vWD. For diagnostic testing, ideally measure vWF off estrogen / in the early follicular phase; if a patient is on OCPs, a normal level does not exclude disease.

BOARD TRAP

WRONG: excluding vWD based on normal vWF levels drawn while the patient is on estrogen-containing OCPs. Discriminator: estrogen falsely raises vWF — repeat testing off hormones (or interpret with caution) when clinical suspicion remains.

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6. Acquired vWS

WHAT YOU SEE

The ACQUIRED VWS panel with a lymphoproliferative/MGUS cell labeled 'PARAPROTEINS BINDING AND CLEARING vWF / MOST COMMON ASSOCIATION' — late-onset acquired syndrome from a clone.

WHAT IT ENCODES

Acquired von Willebrand SYNDROME (AVWS) presents as new-onset mucocutaneous bleeding in an OLDER patient with NO family history. Major mechanisms: (1) autoantibody/adsorption in lymphoproliferative disorders & MGUS, (2) adsorption onto malignant/clonal cells, (3) mechanical shear destruction (aortic stenosis, LVAD, extreme thrombocytosis). Workup the underlying disorder; treat the cause. For acute bleeding: IVIG often helps the MGUS-associated form; high-dose vWF concentrate or recombinant FVIIa for refractory bleeds.

BOARD TRAP

WRONG: labeling a new bleeding tendency in an elderly patient as 'late-onset hereditary vWD.' Discriminator: absence of lifelong/family bleeding history + an associated condition (MGUS, lymphoma, aortic stenosis, thrombocytosis) points to ACQUIRED vWS — search for and treat the cause.

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7. Heyde / shear destruction

WHAT YOU SEE

The heart-and-valve image 'HIGH SHEAR STRESS MECHANICALLY UNFOLDS vWF STRINGS / ADAPTING FROM ANGIODYSPLASIA / GI BLEEDING AFTER VALVE REPLACEMENT' — Heyde shear shredding large multimers.

WHAT IT ENCODES

Heyde syndrome = aortic STENOSIS + GI bleeding from angiodysplasia. High shear across the stenotic valve unfolds vWF and exposes it to ADAMTS13 cleavage, destroying the largest (most hemostatic) multimers → acquired type-2A-like vWS. The same shear mechanism occurs with LVADs. Definitive treatment is AORTIC VALVE REPLACEMENT, which corrects the multimer loss and typically stops the bleeding.

BOARD TRAP

WRONG: managing recurrent GI angiodysplasia bleeding in an aortic-stenosis patient with endoscopic cautery alone and ignoring the valve. Discriminator: in Heyde syndrome, valve REPLACEMENT addresses the root cause (restores large vWF multimers); endoscopic therapy treats only the symptom.

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8. Extreme thrombocytosis

WHAT YOU SEE

A massive pile of platelets stamped '>1.5 MILLION' labeled 'EXTREME THROMBOCYTOSIS / SAME PARADOX AS ET' — too many platelets adsorb vWF and cause bleeding.

WHAT IT ENCODES

Counterintuitively, EXTREME thrombocytosis (platelets >1.0-1.5 million, as in essential thrombocythemia or reactive thrombocytosis) causes BLEEDING, not clotting, via acquired vWS: the huge platelet mass ADSORBS large vWF multimers off the plasma. Treatment is to LOWER the platelet count (cytoreduction, e.g., hydroxyurea ± aspirin caution) — and notably aspirin is risky at these extreme counts.

BOARD TRAP

WRONG: starting/continuing antiplatelet therapy for 'thrombosis prevention' in a patient with platelets >1.5 million who is bleeding. Discriminator: at extreme counts the problem is acquired vWS-mediated BLEEDING — cytoreduce the platelets; do not reflexively add aspirin, which can precipitate hemorrhage.

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9. DDAVP first-line type 1

WHAT YOU SEE

The DDAVP delivery truck stamped 'ALWAYS TRIAL BEFORE PROCEDURE' with 'FIRST-LINE TYPE 1, 0.3 mcg/kg, PEAK 30 MIN' — the desmopressin first-line agent that needs a trial run.

WHAT IT ENCODES

DESMOPRESSIN (DDAVP) is FIRST-LINE for type 1 vWD and mild hemophilia A. It is a synthetic vasopressin (V2) analog that triggers release of stored vWF and FVIII from endothelial Weibel-Palade bodies, raising levels ~2-4x. Dosing: 0.3 mcg/kg IV/SC, or intranasal 150 mcg (1 spray) for <50 kg / 300 mcg (both nostrils) for ≥50 kg; peaks at ~30-60 min. Always do a TRIAL/challenge to confirm an adequate response before relying on it for surgery or bleeds. Add tranexamic acid for mucosal procedures.

BOARD TRAP

WRONG: using DDAVP without first documenting a response trial. Roughly 1 in 4 type 1 patients respond poorly, and type 2/3 patients respond poorly or are harmed. Discriminator: confirm a successful DDAVP challenge (adequate vWF/FVIII rise) before depending on it; if response is inadequate, use vWF concentrate.

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10. DDAVP tachyphylaxis

WHAT YOU SEE

The DDAVP truck note 'TACHYPHYLAXIS / SPACE DOSES 12-24h' plus 'FLUID RESTRICTION 24h, HYPONATREMIA RISK' — stores deplete with repeat dosing.

WHAT IT ENCODES

DDAVP works by depleting endothelial vWF STORES, so repeated dosing shows TACHYPHYLAXIS — the second and later doses release progressively less. Space doses every 12-24 h and limit to ~2-3 doses, then switch to vWF concentrate for ongoing needs. Its V2 antidiuretic effect causes free-water retention → HYPONATREMIA and seizures, so FLUID-RESTRICT and avoid in young children, elderly, and patients with active coronary/vascular disease.

BOARD TRAP

WRONG: continuing DDAVP every several hours for days for a major bleed/surgery. Discriminator: stores deplete (tachyphylaxis) and hyponatremia accrues — after ~2-3 spaced doses switch to vWF concentrate, restrict fluids, and monitor sodium.

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11. DDAVP contraindications

WHAT YOU SEE

The DDAVP COMPATIBILITY SHELF: TYPE 1 green check, TYPE 2B and TYPE 3 red X'd ('CONTRAINDICATED / NO STORES, NO RESPONSE') — desmopressin barred in 2B and 3.

WHAT IT ENCODES

DDAVP is INEFFECTIVE in type 3 (no vWF stores to release) and CONTRAINDICATED in type 2B (releases hyperadhesive vWF → worsens platelet clumping and thrombocytopenia). It is also relatively contraindicated in significant cardiovascular disease (the released vWF/FVIII plus fluid shift add thrombotic and volume risk). Use vWF-containing concentrate instead in these settings.

BOARD TRAP

WRONG: reaching for desmopressin in a vWD patient who has thrombocytopenia + enhanced low-dose ristocetin (type 2B) or near-absent vWF (type 3). Discriminator: type 2B and type 3 → vWF concentrate, never DDAVP.

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12. vWF replacement

WHAT YOU SEE

The vWF REPLACEMENT SHELF of concentrate bottles 'vWF CONCENTRATE NOT JUST FVIII / INFUSED FVIII RAPIDLY CLEARED WITHOUT FUNCTIONAL CARRIER' — the factor concentrate for type 3 / DDAVP non-responders.

WHAT IT ENCODES

vWF-containing CONCENTRATE is the treatment for type 3, severe type 2, type 2B, DDAVP non-responders, and major bleeding/surgery. Options: plasma-derived vWF/FVIII concentrate (e.g., Humate-P/Wilate) or recombinant vWF (vonicog alfa, contains no FVIII so a first FVIII dose is co-given when FVIII is low). Dose and monitor by vWF:RCo (and FVIII) units, targeting hemostatic levels through the bleeding/surgical period.

BOARD TRAP

WRONG: substituting cryoprecipitate or fresh frozen plasma as the modern standard for vWD. Discriminator: virus-inactivated vWF/FVIII concentrate (or recombinant vWF) is preferred; cryoprecipitate is a last-resort/legacy option because it is not virally inactivated.

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13. Antifibrinolytics adjunct

WHAT YOU SEE

The ANTIFIBRINOLYTIC shelf with TXA and EACA bottles, good for menstrual/dental bleeding but 'CONTRAINDICATED IN UPPER URINARY TRACT BLEEDING / URETERAL OBSTRUCTION' — clot-stabilizing mucosal adjuncts.

WHAT IT ENCODES

Antifibrinolytics — TRANEXAMIC ACID (TXA) and aminocaproic acid — stabilize formed clot by blocking plasminogen activation. They are excellent ADJUNCTS for MUCOSAL bleeding: menorrhagia, epistaxis, and dental/oral procedures (oral rinse). TXA dosing is typically 1 g (or ~10-15 mg/kg) PO/IV q6-8h. They are also estrogen-free options for menorrhagia. Patients with vWD should AVOID aspirin/NSAIDs, which further impair platelet function.

BOARD TRAP

WRONG: giving tranexamic acid for UPPER URINARY TRACT (renal/ureteral) bleeding. Antifibrinolytics promote clot retention in the collecting system → ureteral obstruction/clot colic. Discriminator: TXA is contraindicated in upper urinary tract bleeding; use it for mucosal/menstrual/dental bleeding instead.